

NumPy Example 1

Creating a Rank 1 Array

```
import numpy as np

# Create a rank 1 array
a = np.array([0, 1, 2])

print(type(a))
print(a.shape)
print(a[0])
print(a[1])
print(a[2])

# Change an element
a[0] = 5

print(a)
```

```
<class 'numpy.ndarray'>
(3,)
0
1
2
[5 1 2]
```

NumPy Example 2

Creating Different Types of Arrays

```
import numpy as np

# Create a 3x3 array of all zeros
a = np.zeros((3,3))
print(a)

# Create a 2x2 array of all ones
b = np.ones((2,2))
print(b)

# Create a 3x3 constant array
c = np.full((3,3), 7)
print(c)

# Create a 3x3 array filled with random values
d = np.random.random((3,3))
print(d)

# Create a 3x3 identity matrix
e = np.eye(3)
print(e)

# Convert list to array
f = np.array([2,3,1,0])
print(f)

# Create an array with values from 0 to 19
g = np.arange(20)
print(g)

# Mixed data types
h = np.array([[0,1,2.0],[0,0,0]],[1+1j,3.,2.]))
print(h)

# Create an array of floats
i = np.arange(1,8,dtype=float)
print(i)

# Create 5 equally spaced numbers between 2 and 4
j = np.linspace(2,4,5)
print(j)
```

```

[[0. 0. 0.]
 [0. 0. 0.]
 [0. 0. 0.]]
[[1. 1.]
 [1. 1.]]
[[7 7 7]
 [7 7 7]
 [7 7 7]]
[[0.60568784 0.31015776 0.96052722]
 [0.13187013 0.31066918 0.78294371]
 [0.75190773 0.58681153 0.29773781]]
[[1. 0. 0.]
 [0. 1. 0.]
 [0. 0. 1.]]
[[2 3 1 0]
 [ 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19]
 [[0.+0.j 1.+0.j 2.+0.j]
 [0.+0.j 0.+0.j 0.+0.j]
 [1.+1.j 3.+0.j 2.+0.j]]
[1. 2. 3. 4. 5. 6. 7.]
[2. 2.5 3. 3.5 4. ]

```

NumPy Example 3

NumPy Data Types

```

import numpy as np

# Let NumPy choose the data type
x = np.array([0, 1])
y = np.array([2.0, 3.0])

# Force a particular data type
z = np.array([5, 6], dtype=np.int64)

print("x dtype:", x.dtype)
print("y dtype:", y.dtype)
print("z dtype:", z.dtype)

x dtype: int64
y dtype: float64
z dtype: int64

```

NumPy Example 4

Array Indexing

```

import numpy as np

# An example array
arr = np.array([
    [-1, 2, 0, 4],
    [4, -0.5, 6, 0],
    [2.6, 0, 7, 8],
    [3, -7, 4, 2.0]
])

# Slicing array
temp = arr[:2, ::2]
print("Array with first 2 rows and alternate columns (0 and 2):\n", temp)

# Integer array indexing
temp = arr[[0, 1, 2, 3], [3, 2, 1, 0]]
print("\nElements at indices (0,3), (1,2), (2,1), (3,0):\n", temp)

# Boolean array indexing
cond = arr > 0
temp = arr[cond]
print("\nElements greater than 0:\n", temp)

Array with first 2 rows and alternate columns (0 and 2):
[[ -1.  0.]
 [ 4.  6.]]

Elements at indices (0,3), (1,2), (2,1), (3,0):
[4. 6. 0. 3.]

```

```
Elements greater than 0:
[2.  4.  4.  6.  2.6 7.  8.  3.  4.  2. ]
```

NumPy Example 5

Basic Array Operations

```
import numpy as np

# Python program to demonstrate
# basic operations on single array

a = np.array([1, 2, 5, 3])

# Add 1 to every element
print("Adding 1 to every element:", a + 1)

# Subtract 3 from each element
print("Subtracting 3 from each element:", a - 3)

# Multiply each element by 10
print("Multiplying each element by 10:", a * 10)

# Square each element
print("Squaring each element:", a ** 2)

# Modify existing array
a *= 2
print("Doubled each element of original array:", a)

# Transpose of array
a = np.array([[1, 2, 3],
              [3, 4, 5],
              [9, 6, 0]])

print("\nOriginal array:\n", a)
print("Transpose of array:\n", a.T)

Adding 1 to every element: [2 3 6 4]
Subtracting 3 from each element: [-2 -1  2  0]
Multiplying each element by 10: [10 20 50 30]
Squaring each element: [ 1  4 25  9]
Doubled each element of original array: [ 2  4 10  6]

Original array:
[[1 2 3]
 [3 4 5]
 [9 6 0]]
Transpose of array:
[[1 3 9]
 [2 4 6]
 [3 5 0]]
```

NumPy Example 6

Unary Operators

```
import numpy as np

# Python program to demonstrate unary operators in NumPy

arr = np.array([
    [1, 5, 6],
    [4, 7, 2],
    [3, 1, 9]
])

# Maximum element of array
print("Largest element is:", arr.max())

# Row-wise maximum elements
print("Row-wise maximum elements:", arr.max(axis=1))

# Minimum element of array (column-wise)
print("Column-wise minimum elements:", arr.min(axis=0))
```

```
# Sum of all array elements
print("Sum of all array elements:", arr.sum())

# Cumulative sum along each row
print("Cumulative sum along each row:\n", arr.cumsum(axis=1))

Largest element is: 9
Row-wise maximum elements: [6 7 9]
Column-wise minimum elements: [1 1 2]
Sum of all array elements: 38
Cumulative sum along each row:
[[ 1  6 12]
 [ 4 11 13]
 [ 3  4 13]]
```

NumPy Example 7

Binary Operators

```
import numpy as np

# Python program to demonstrate
# binary operators in NumPy

a = np.array([[1, 2],
              [3, 4]])

b = np.array([[4, 3],
              [2, 1]])

# Add arrays
print("Array sum:\n", a + b)

# Multiply arrays (element-wise multiplication)
print("\nArray multiplication:\n", a * b)

# Matrix multiplication
print("\nMatrix multiplication:\n", a.dot(b))

Array sum:
[[5 5]
 [5 5]]

Array multiplication:
[[4 6]
 [6 4]]

Matrix multiplication:
[[ 8  5]
 [20 13]]
```

NumPy Example 8

Universal Functions (ufunc)

```
import numpy as np

# Python program to demonstrate
# universal functions in NumPy

# Create an array of sine values
a = np.array([0, np.pi/2, np.pi])
print("Sine values of array elements:", np.sin(a))

# Exponential values
a = np.array([0, 1, 2, 3])
print("Exponent of array elements:", np.exp(a))

# Square root values
print("Square root of array elements:", np.sqrt(a))

Sine values of array elements: [0.0000000e+00 1.0000000e+00 1.2246468e-16]
Exponent of array elements: [ 1.          2.71828183  7.3890561  20.08553692]
Square root of array elements: [0.          1.          1.41421356  1.73205081]
```



NumPy Example 9

Matrix Inverse

```
import numpy as np

# Python program to find the inverse of a matrix

arr = np.array([
    [-1, 2, 0, 4],
    [4, -0.5, 6, 0],
    [2.6, 0, 7, 8],
    [3, -7, 4, 2.0]
])

inverse = np.linalg.inv(arr)

print("Inverse of matrix:")
print(inverse)
```

```
Inverse of matrix:
[[ 4.05707196  1.98511166 -2.28287841  1.01736973]
 [ 0.50620347  0.30272953 -0.24813896 -0.01985112]
 [-2.66253102 -1.13151365  1.50124069 -0.67990074]
 [ 1.01116625  0.34491315 -0.44665012  0.26426799]]
```

Pandas Example 1

Creating a Series

```
import pandas as pd
import numpy as np

# Creating a Series
s = pd.Series(
    [1, 2, 3, np.nan, 5, 6],
    index=['A', 'B', 'C', 'D', 'E', 'F']
)

print(s)
```

```
A    1.0
B    2.0
C    3.0
D    NaN
E    5.0
F    6.0
dtype: float64
```

Pandas Example 2

Creating a DataFrame

```
import pandas as pd

data = {
    'Gender': ['F', 'M', 'M'],
    'Emp_ID': ['E01', 'E02', 'E03'],
    'Age': [25, 27, 25]
}

# Specify the column order
df = pd.DataFrame(data, columns=['Emp_ID', 'Gender', 'Age'])

print(df)          # Print the entire DataFrame
print(df.Emp_ID)   # Print the Emp_ID column
print(df.Emp_ID[0]) # Print the first Emp_ID
```

```
   Emp_ID Gender  Age
0    E01      F   25
1    E02      M   27
2    E03      M   25
0    E01
1    E02
2    E03
```

```
Name: Emp_ID, dtype: object
E01
```

▼ Pandas Example 3

Reading CSV Files

```
import pandas as pd

# Read the mtcars dataset
df = pd.read_csv("mtcars.csv")

# Display the first 5 rows
print(df.head())
```

	model	mpg	cyl	disp	hp	drat	wt	qsec	vs	am	gear	\
0	Mazda RX4	21.0	6	160.0	110	3.90	2.620	16.46	0	1	4	
1	Mazda RX4 Wag	21.0	6	160.0	110	3.90	2.875	17.02	0	1	4	
2	Datsun 710	22.8	4	108.0	93	3.85	2.320	18.61	1	1	4	
3	Hornet 4 Drive	21.4	6	258.0	110	3.08	3.215	19.44	1	0	3	
4	Hornet Sportabout	18.7	8	360.0	175	3.15	3.440	17.02	0	0	3	

	carb
0	4
1	4
2	1
3	1
4	2

▼ Pandas Example 4

Descriptive Statistics

```
import pandas as pd

# Read Iris dataset
iris = pd.read_csv("Iris.csv")

# Display descriptive statistics
print(iris.describe())
```

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm
count	150.000000	150.000000	150.000000	150.000000	150.000000
mean	75.500000	5.843333	3.054000	3.758667	1.198667
std	43.445368	0.828066	0.433594	1.764420	0.763161
min	1.000000	4.300000	2.000000	1.000000	0.100000
25%	38.250000	5.100000	2.800000	1.600000	0.300000
50%	75.500000	5.800000	3.000000	4.350000	1.300000
75%	112.750000	6.400000	3.300000	5.100000	1.800000
max	150.000000	7.900000	4.400000	6.900000	2.500000

▼ Pandas Example 5

Covariance Matrix

```
import pandas as pd

# Read Iris dataset
iris = pd.read_csv("Iris.csv")

# Calculate covariance matrix
print(iris.cov(numeric_only=True))
```

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	\
Id	1887.500000	25.782886	-7.492282	67.667785	
SepalLengthCm	25.782886	0.685694	-0.039268	1.273682	
SepalWidthCm	-7.492282	-0.039268	0.188004	-0.321713	
PetalLengthCm	67.667785	1.273682	-0.321713	3.113179	
PetalWidthCm	29.832215	0.516904	-0.117981	1.296387	

	PetalWidthCm
Id	29.832215
SepalLengthCm	0.516904
SepalWidthCm	-0.117981
PetalLengthCm	1.296387
PetalWidthCm	0.582414

✓ Pandas Example 6

Correlation **Matrix**

```
import pandas as pd

# Read Iris dataset
iris = pd.read_csv("Iris.csv")

# Calculate correlation matrix
print(iris.corr(numeric_only=True))
```

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	\
Id	1.000000	0.716676	-0.397729	0.882747	
SepalLengthCm	0.716676	1.000000	-0.109369	0.871754	
SepalWidthCm	-0.397729	-0.109369	1.000000	-0.420516	
PetalLengthCm	0.882747	0.871754	-0.420516	1.000000	
PetalWidthCm	0.899759	0.817954	-0.356544	0.962757	

	PetalWidthCm
Id	0.899759
SepalLengthCm	0.817954
SepalWidthCm	-0.356544
PetalLengthCm	0.962757
PetalWidthCm	1.000000